EMS611U: Conceptual Design of Aircraft

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Sem A - Week 4

Road Map

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Office Hours:

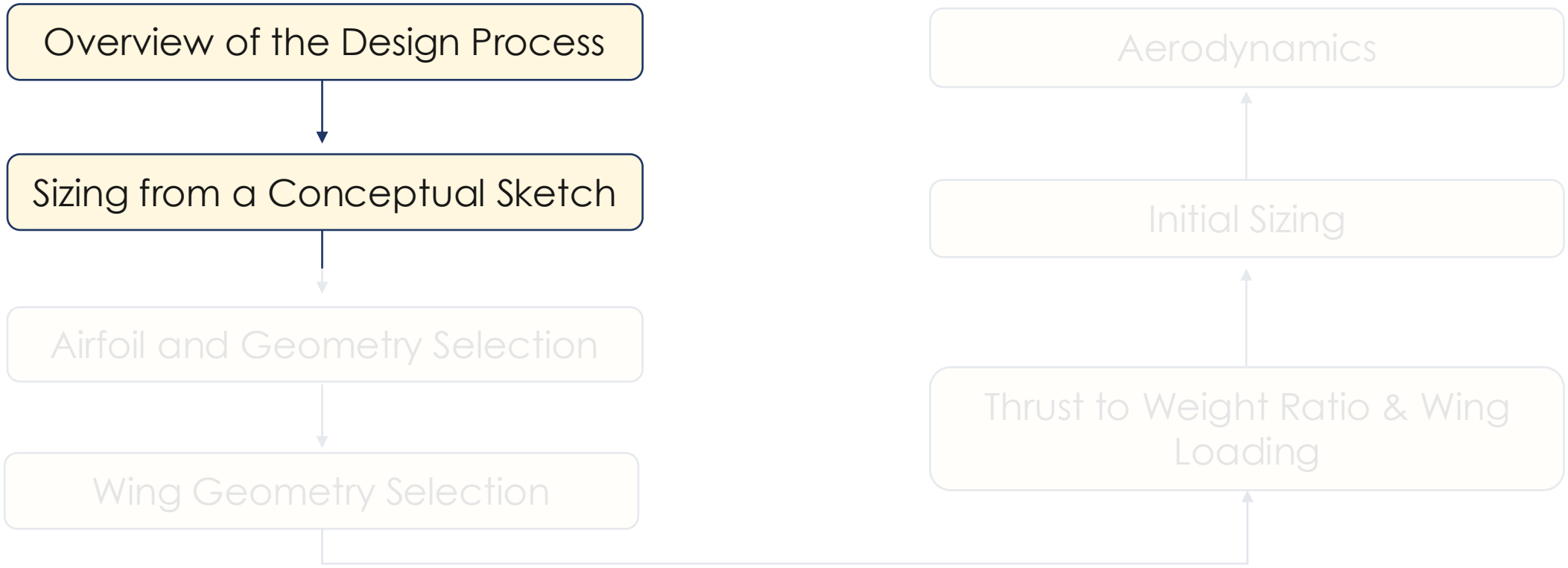
- Wednesday 9:00 am – 10:00 am (in person)
- Wednesday 10:00 am – 11:00 am (online)

Coursebooks – Available in Library:

1. Raymer D. P, 2012, Aircraft Design: A Conceptual Approach, Fourth Edition, AIAA Education Series, New York, NY, ISBN:1-56347-829.
2. Roskam, Jan, & Lan, C. Edward. Airplane Aerodynamics and Performance. Lawrence, KS: DARcorporation, 1997. ISBN: 1884885446.
3. Kundu, Ajoy Kumar. Aircraft Design. Cambridge Aerospace Series, Volume 27. Cambridge University Press, 2010. ISBN: 9781139487450.

Road Map

What will we be focusing on in this portion of the module ?



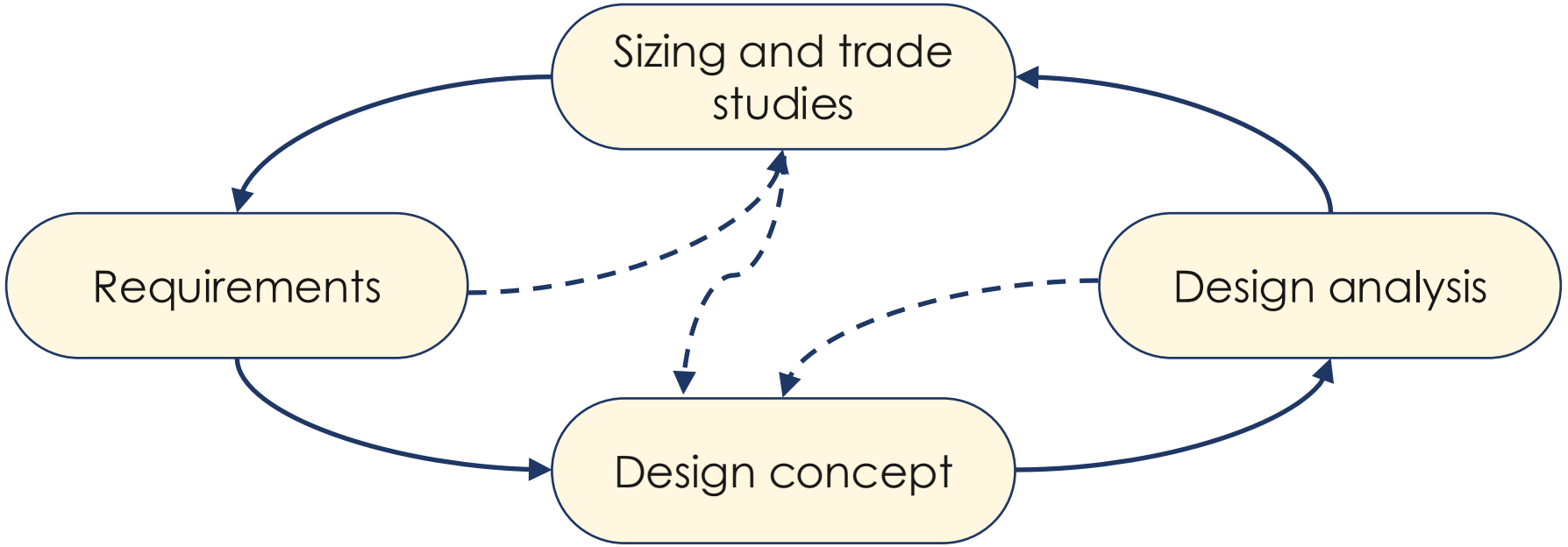
Overview of the Design Process

Analytical disciplines in aerospace Engineering:

- Aerodynamics, structures, controls, propulsion

What is design?

- Creating the geometric description of thing to be built.



- Design is an *iterative effort*.
- Concepts are developed to meet the requirements.

Overview of the Design Process

Requirements

↓

Conceptual Design

- Will it work?
- What does it look like?
- What requirements drive the design?
- What trade-offs should be considered?
- What should it weigh and cost?

Preliminary Design

- Freeze the configuration
- Develop test and analytical base
- Design major items
- Develop actual cost estimate

Detail Design

- Design the actual pieces to be built
- Design the tooling and fabrication process
- Test major items
- Finalize weight and performance estimates

↓

Fabrication

Sizing from a Conceptual Sketch

What do we utilise whilst we are designing an aircraft?

Imagine you have been hired as an aerospace engineer, and you have been tasked to design a fighter to replace Air Force F-15 and you need an immediate estimate of its take-off weight. What would you do?

- The simplest level just adopts **past history**.

- To get a design "right":
  - Several years
  - Many people
  - Lots of money
  - Lots of candidate designs
  - Analysis techniques (e.g. Codes, wind tunnel tests etc.)

PS: Even with this much of effort, the aeroplane when flown will never exactly match predictions.

Where do we start and how do we start?

- Conceptual design activities.
  - The simplified sizing method (not suitable for combat or payload drops).
  - It is crude but this method introduces all of the essential features of the most sophisticated design by major companies.

Takeoff-Weight Build up

W_0: Design Takeoff Weight Gross Weight

- The total weight of the aircraft as it begins the mission for which it was designed.
- This is not necessarily same as the "maximum takeoff weight".
- Many military aircraft can be overloaded beyond design weight but will suffer from a reduced maneuverability.

$$W_0 = W_{crew} + W_{payload} + W_{fuel} + W_{empty}$$

W_{empty}: Structures, engines, landing gears, fixed equipment, avionics, etc.

Known weights:
Design requirements

Unknown weights:
Depens on total aircraft weight

An iterative process must be used for aircraft sizing.

Takeoff-Weight Build up

$$W_0 = W_{crew} + W_{payload} + W_{fuel} + W_{empty}$$

$$W_0 = W_{crew} + W_{payload} + \left(\frac{W_f}{W_0} W_0\right) + \left(\frac{W_e}{W_0} W_o\right)$$

$$W_0 - \left(\frac{W_f}{W_0}\right) W_0 - \left(\frac{W_e}{W_0}\right) W_o = W_{crew} + W_{payload}$$

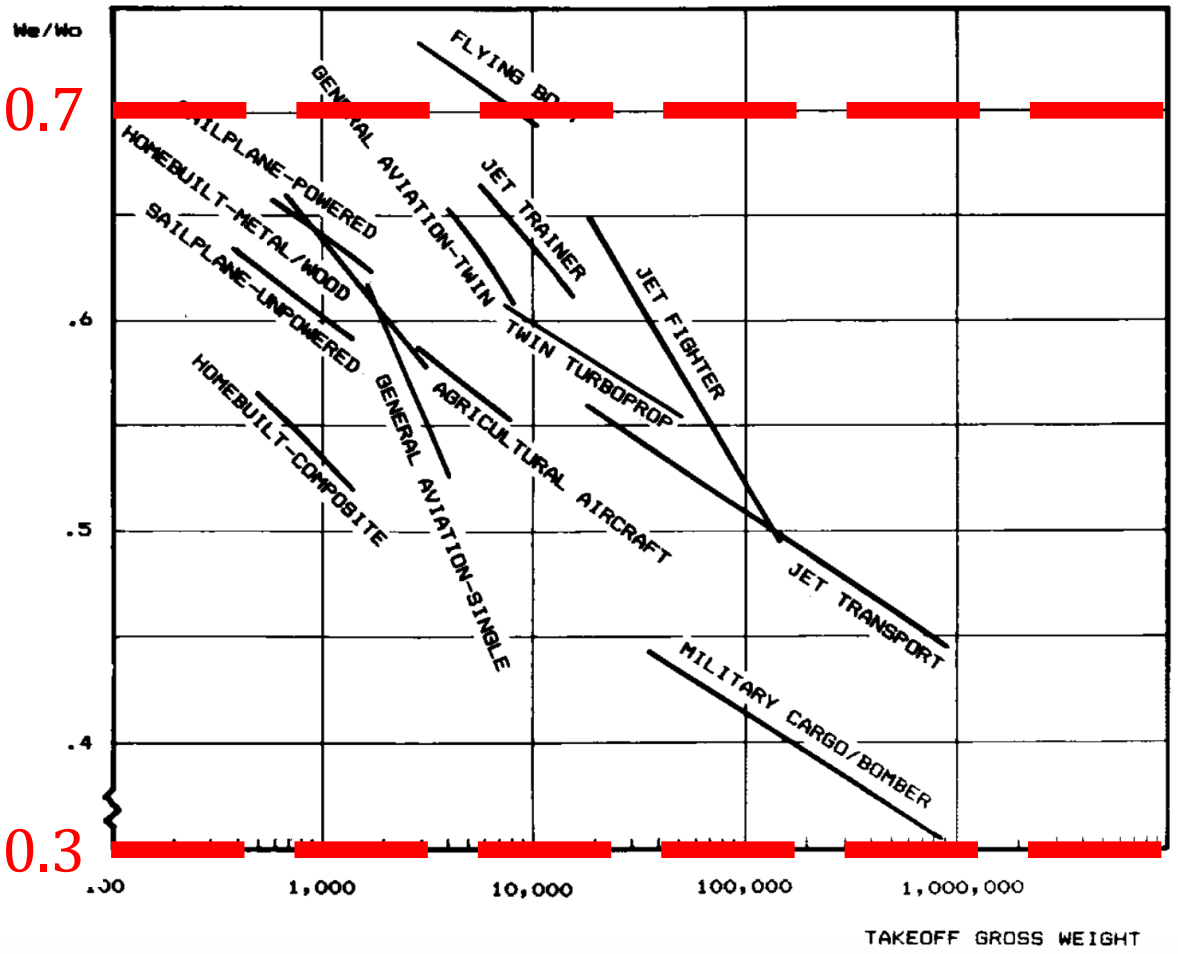
$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$

Requirement parameters

can be estimated

Empty Weight Fraction (W_e/W_0) Estimation

- The empty weight fraction can be estimated statistically from historical trends.



- Empty weight fractions vary from about 0.3 to 0.7 and diminish with increasing total weight.

$$W_0 \uparrow \rightarrow \frac{W_e}{W_0} \downarrow$$

Empty Weight Fraction (W_e/W_0) Estimation

The empty weight fraction coefficients (statistical curve-fit equations for the trends)

	A	C
Sailplane—unpowered	0.86	−0.05
Sailplane—powered	0.91	−0.05
Homebuilt—metal/wood	1.19	−0.09
Homebuilt—composite	0.99	−0.09
General aviation—single engine	2.36	−0.18
General aviation—twin engine	1.51	−0.10
Agricultural aircraft	0.74	−0.03
Twin turboprop	0.96	−0.05
Flying boat	1.09	−0.05
Jet trainer	1.59	−0.10
Jet fighter	2.34	−0.13
Military cargo/bomber	0.93	−0.07
Jet transport	1.02	−0.06

$$\frac{W_e}{W_0} = A\, W_0^{C} K_{vs}$$



K_{vs} = variable sweep constant = 1.04 if variable sweep
= 1.00 if fixed sweep

Fuel fraction (W_f/W_0) Estimation

- The majority of the aircraft's fuel supply is available for performing the mission.

- The other fuel includes "reserve fuel" as required by civil or military design specifications and also includes "trapped fuel" which is the fuel that cannot be pumped out of the tanks, generally assumed to be around 6%.

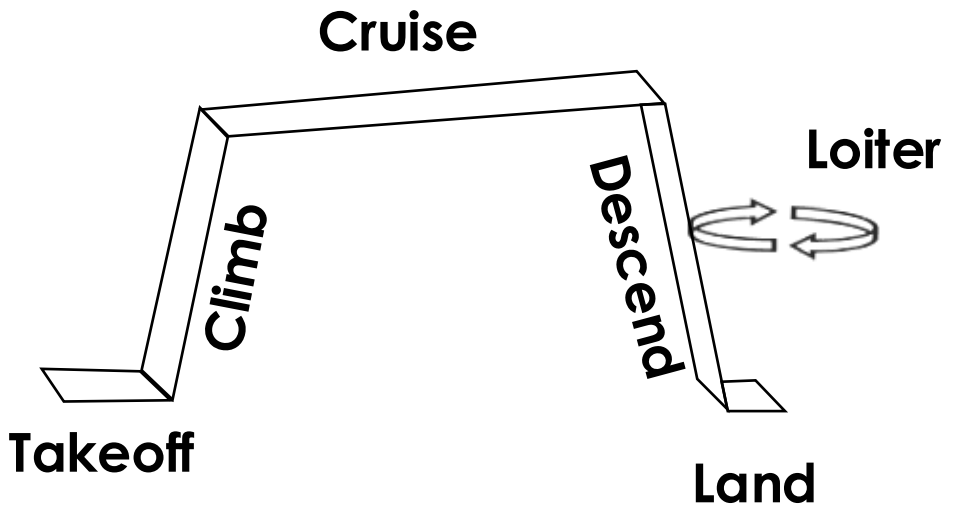
The required amount of **mission fuel depends on:**

- The mission to be flown
- The aerodynamics of the aircraft
- The engines fuel consumption

The aircraft weight during the mission affects the drag, so the fuel used is a function of the aircraft weight.

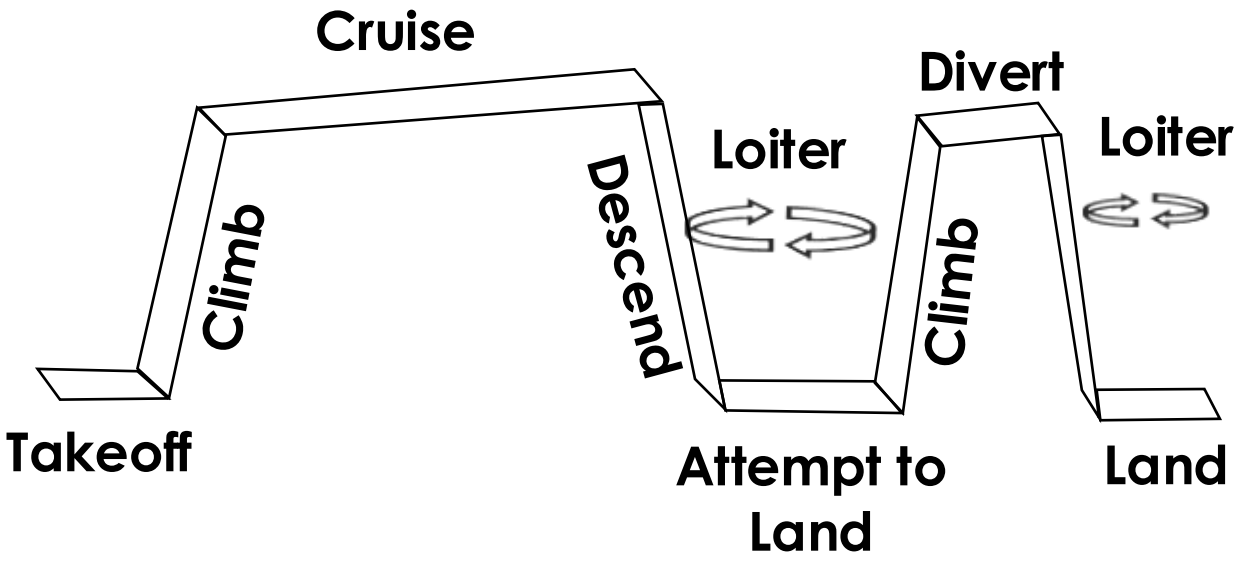
Fuel fraction, W_f/W_0, can be estimated based on the mission to be flown using approximation of the fuel consumption and aerodynamics.

Simple Cruise Mission:



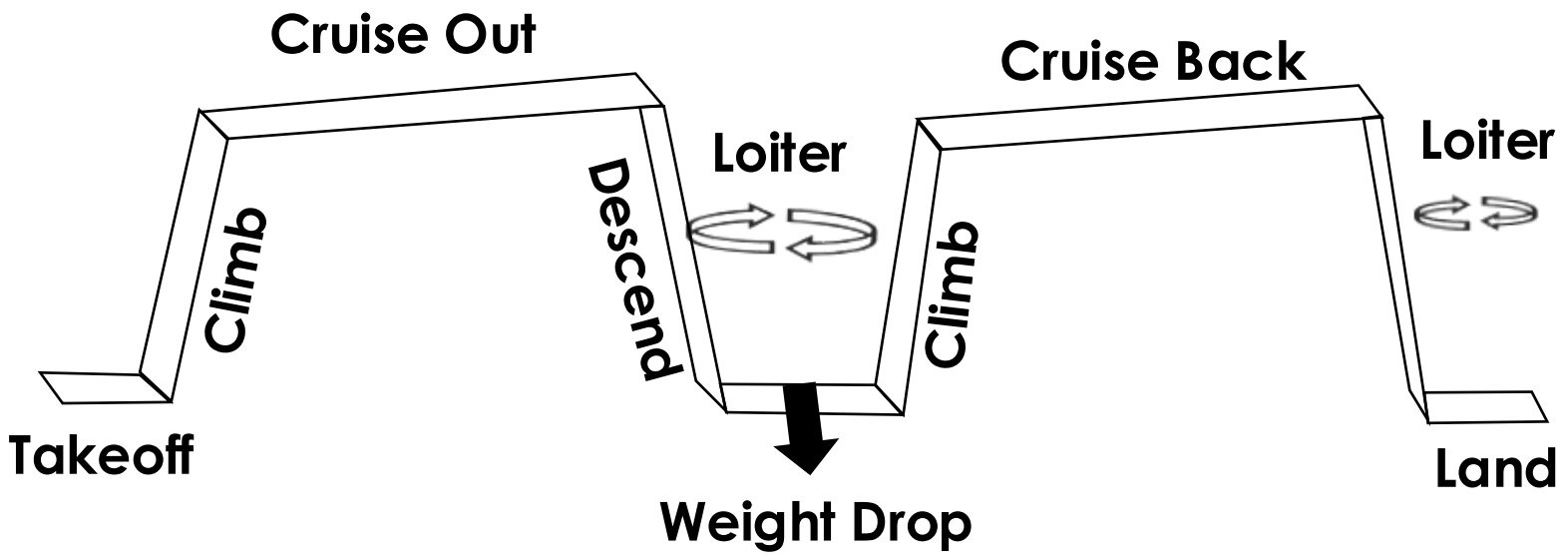
- Simple cruise mission is used for many transport and general-aviation designs, including homebuilts.
- The aircraft is sized to provide some required cruise range.
- For safety it would be wise to carry extra fuel in case your intended airport is closed, so a loiter of typically 20-30 minutes (at 3048 m altitude) is added.

Commercial Transport:



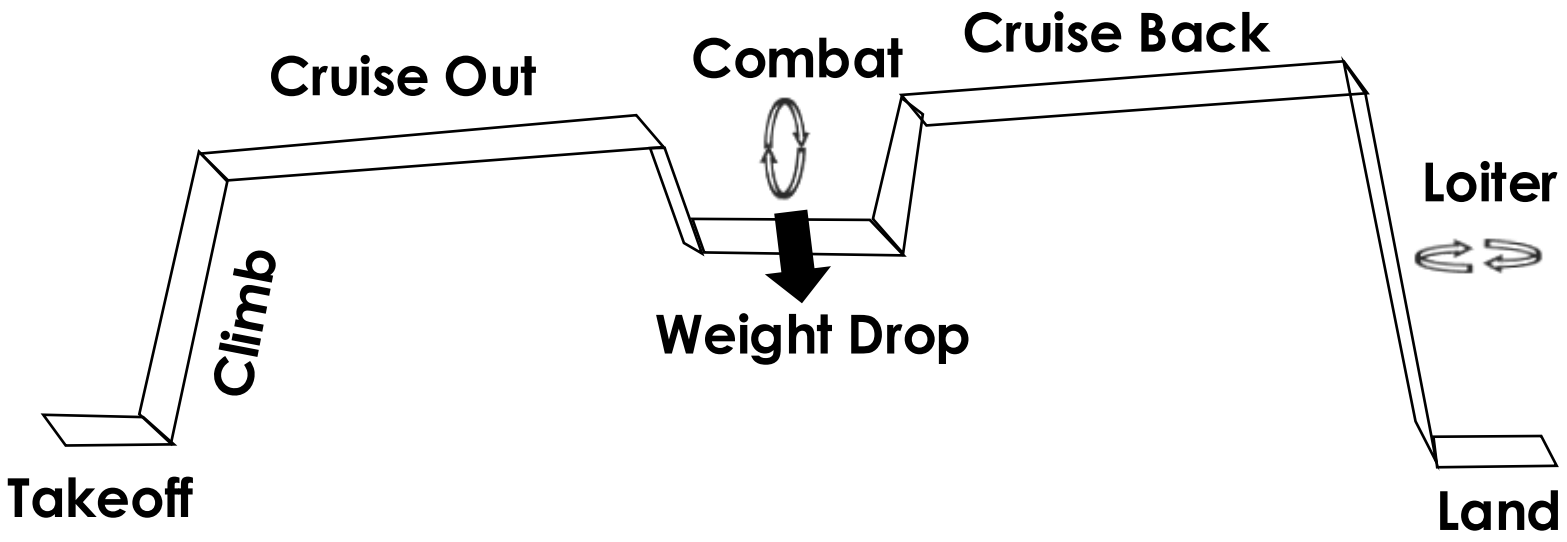
- Additional range could be included, representing the distance to the nearest other airport or some fixed number of minutes of flight at cruise speed.

Low-level Strike Mission:



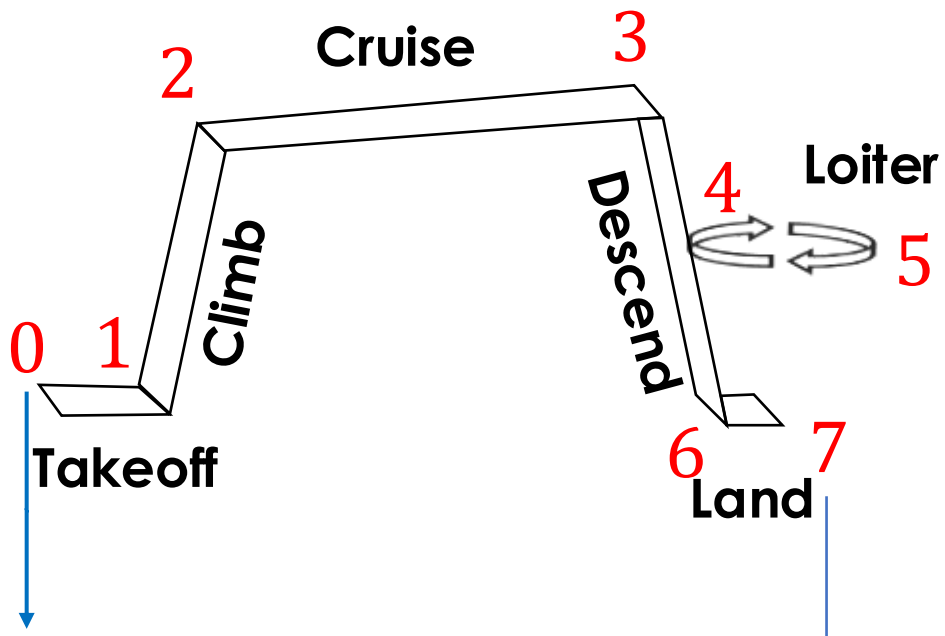
- The low-level strike mission includes "dash" segments that must be flown at just a few hundred feet off the ground. This is to improve the survivability of the as it approaches its target. The aerodynamic efficiency of an aircraft is greatly reduced during low-level, high-speed flight, as its engine efficiency.

Air-superiority Mission:



- The typical "air superiority mission" includes a cruise out, a combat consisting of either a certain number of turns or a certain number of minutes at maximum power, a weapons drop, a cruise back, and a loiter. The weapons drop refers to the firing of gun and missiles and is often left out of the sizing analysis to ensure that the aircraft has enough fuel to return safely if the weapons are not used. The aircraft must return to its base at the end of the mission.

Mission Segment Weight Fractions



Simple Cruise Mission:

- Warmup and takeoff
- Climb
- Cruise
- Descend
- Loiter
- Land

W_0: Takeoff Gross Weight

W_7: Final Weight

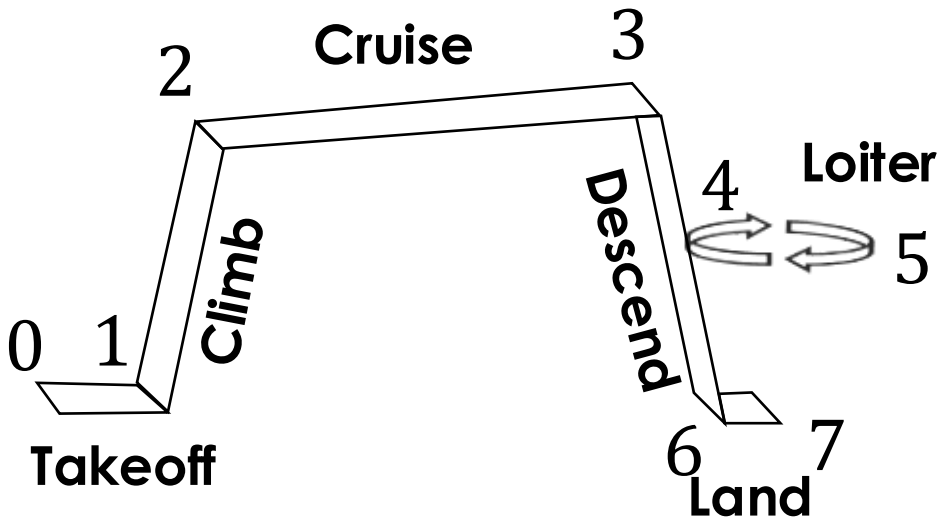
$\frac{W_i}{W_{i-1}}$: Mission segment weight fraction

$\frac{W_x}{W_0}$: Total fuel fraction required

$$\frac{W_{final}}{W_{initial}} = \frac{W_7}{W_0} = \frac{W_1}{W_0}\frac{W_2}{W_1}\frac{W_3}{W_2}\frac{W_4}{W_3}\frac{W_5}{W_4}\frac{W_6}{W_5}\frac{W_7}{W_6}$$

$$W_{initial} - W_{final} = W_{fuel}$$

Mission Segment Weight Fractions



- Takeoff, climb, and landing weight fractions can be estimated from historical data(Averaged values given in the table are reasonable for initial sizing):

	(W_i/W_{i-1})
Warmup and takeoff	0.970
Climb	0.985
Landing	0.995

$$\frac{W_{final}}{W_{initial}} = \frac{W_7}{W_0} = \frac{W_1}{W_0}\frac{W_2}{W_1}\frac{W_3}{W_2}\frac{W_4}{W_3}\frac{W_5}{W_4}\frac{W_6}{W_5}\frac{W_7}{W_6}$$

Takeoff Climb Cruise Descend Loiter Descend Landing

0.97 0.985 0.995

Cruise? Loiter?

$\frac{W_3}{W_2}$ $\frac{W_5}{W_4}$

Mission Segment Weight Fractions

❖ In simple sizing method: the distance traveled during descent is part of the cruise range.

Cruise

$\frac{W_3}{W_2}$

Cruise-segment mission weight fractions can be found using the Breguet range equation:

$$R = \frac{V}{C}\frac{L}{D}\left[ln\left(\frac{W_{i-1}}{W_i}\right)\right] \Longrightarrow \frac{W_i}{W_{i-1}} = e^{\frac{-RC}{V(L/D)}}$$

R: Range
C: Specific fuel consumption
V: Velocity
L/D: Lift-to-drag ratio

Loiter

$\frac{W_5}{W_4}$

Loiter weight fractions are found from the endurance equation:

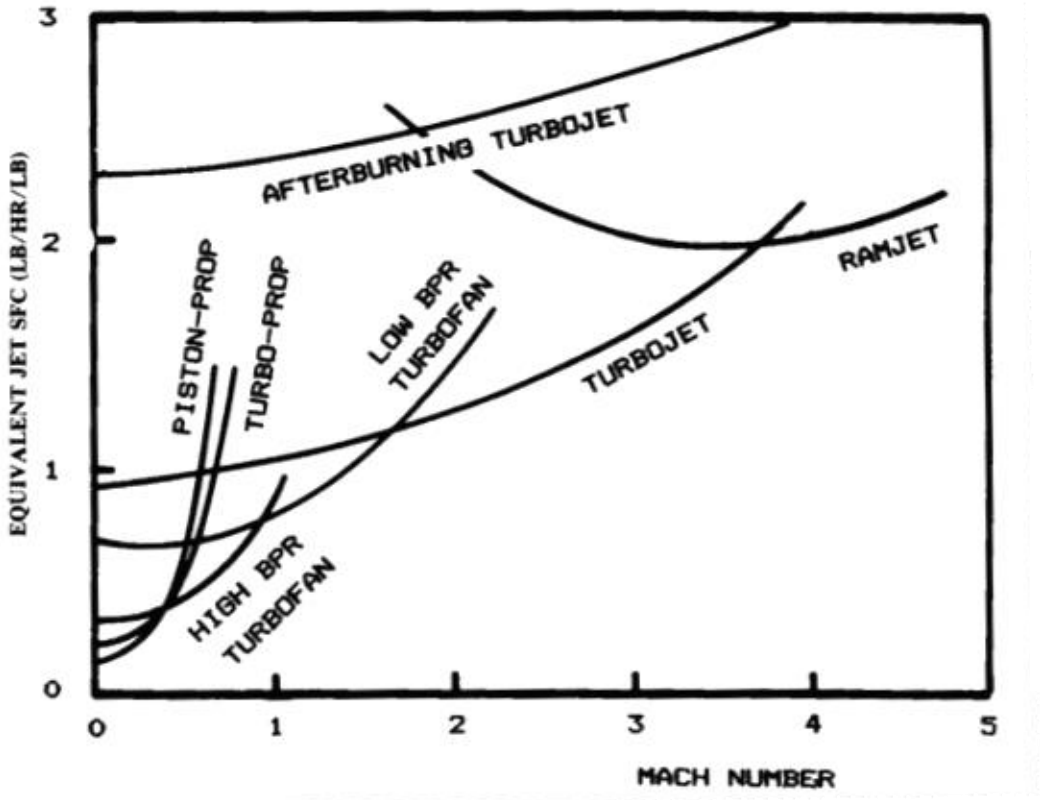
$$E = \frac{1}{C}\frac{L}{D}\left[ln\left(\frac{W_{i-1}}{W_i}\right)\right] \Longrightarrow \frac{W_i}{W_{i-1}} = e^{\frac{-EC}{L/D}}$$

E: Endurance, loiter time

Specific Fuel Consumption, SFC or C

SFC is a measure of the fuel consumed by an engine.

- Jet engines convert the fuel-flow directly into thrust.
- Turbo-prop and piston engines first convert fuel-flow into power output on a shaft. The propeller then takes the power xand converts it into thrust.



Typical jet SFC's	Cruise	Loiter
Pure turbojet	0.9	0.8
Low-bypass turbofan	0.8	0.7
High-bypass turbofan	0.5	0.4

[ft/s]

Propeller: $C = C_{bhp} V / (550 \eta_p)$

Typical C_{bhp} and η_p	Cruise	Loiter
Piston-prop (fixed pitch)	0.4/0.8	0.5/0.7
Piston-prop (variable pitch)	0.4/0.8	0.5/0.8
Turboprop	0.5/0.8	0.6/0.8

C, [1/hr]

$(L/D)_{max}$ Estimation

- The remaining unknown in both range and loiter equations is the L/D, Lift-to-drag ratio.

$$\frac{W_i}{W_{i-1}} = e^{\frac{-RC}{V(L/D)}}$$

$$\frac{W_i}{W_{i-1}} = e^{\frac{-EC}{L/D}}$$

- L/D is highly dependent upon the configuration arrangement.
- At subsonic speeds $\boldsymbol{L/D}$ is most directly affected by **two aspects of the design**:
  - Wing Span
  - Wetted Area
- **In Level Flight:** $\boldsymbol{L = W}$; $\boldsymbol{L/D}$ is solely dependent in D.

$$C_D = \underbrace{C_{D_0}}_{\text{Parasite Drag}} + \underbrace{\frac{C_L^2}{\pi AR\, e}}_{\text{Induced Drag}}$$

Parasite Drag: Primarily caused by skin friction ~ *"wetted area"*

Induced Drag: caused by the generation of lift ~ *span*

- The "aspect ratio" of the wing has historically been used as the primary indicator of wing efficiency.

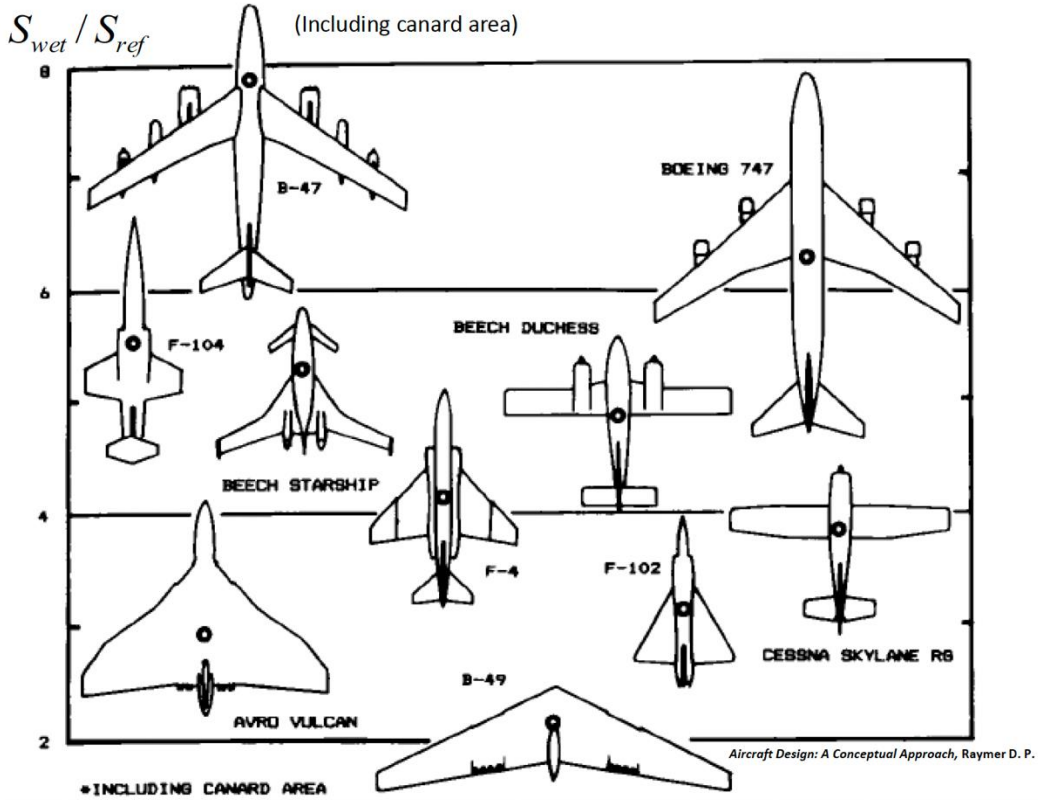
For rectangular wings: $AR = \frac{b}{\bar{c}}$

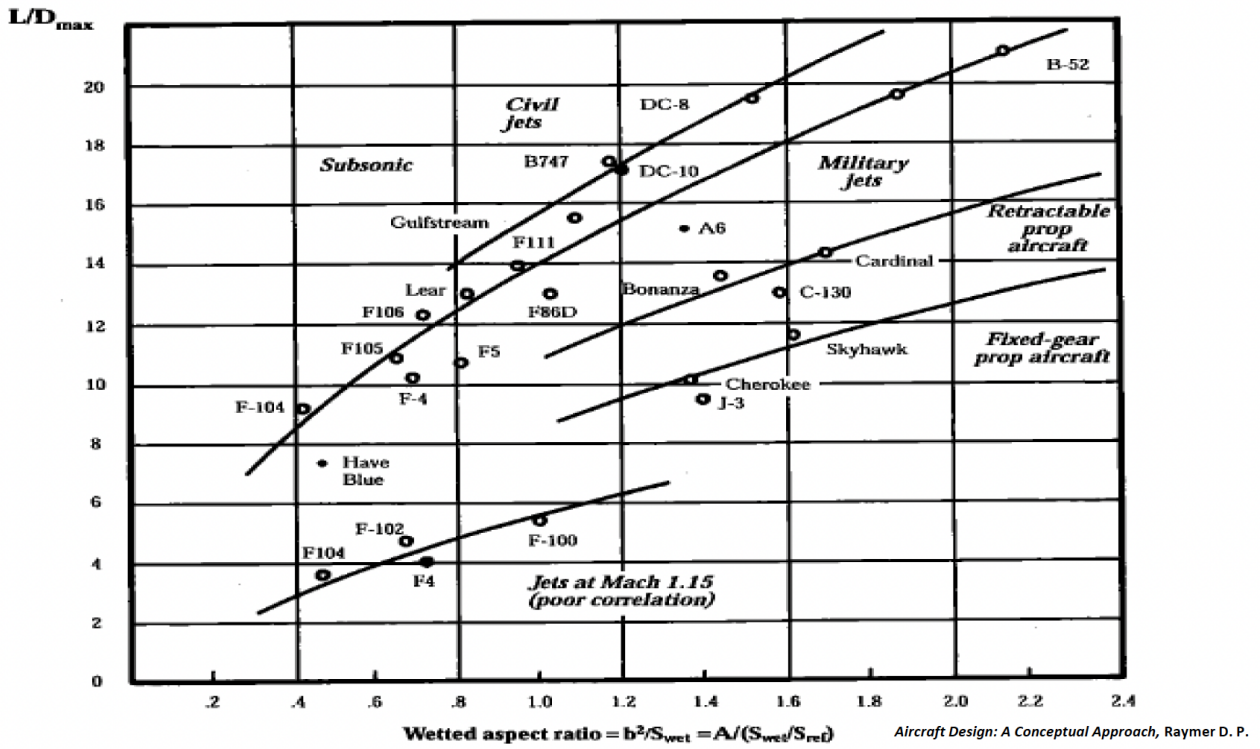
$$AR = \frac{b^2}{S}$$

- Typical values range between 3 and 8.
- For initial design purposes, aspect ratio can be selected from historical data.

$(L/D)_{max}$ Estimation

Wetted-Area Ratio: The wetted-area ratio can be used, along with aspect ratio, as a more reliable early estimate of L/D.





Wetted-Aspect Ratio: $\frac{b^2}{S_{wet}} = \frac{AR}{S_{wet}/S_{ref}}$

$(L/D)_{max}$ Estimation

Notes:

- The designer picks the aspect ratio and determines the configuration arrangement, which in turn determines the wetted-area ratio.
- The designer must strike a compromise between the desire for a high L/D and the conflicting desire for low weight.
- If the aspect ratio selected is much higher than that of other aircraft in its class, the empty weight fraction would be higher than estimated by these simple statistical equations.
- L/D can now be estimated from a conceptual sketch: On the conceptual sketch the designer arranges the major components of the aircraft, including wings, tails, fuselage, engines, payload or passenger compartment, landing gear, fuel tanks, and others as needed.
- From the sketch wetted-area ratio can be estimated.
- The wetted aspect ratio can then be calculated as the wing aspect ratio divided by the wetted-area ratio.
- (L/D)maxgraphic then be used to estimate the maximum L/D.

$(L/D)_{max}$ Estimation

- Drag varies with altitude and velocity. For any altitude there is a velocity that maximizes L/D.
- To maximize cruise or loiter efficiency the aircraft should fly at approximately the velocity for maximum L/D.

	Cruise	Loiter
Jet	$0.866\,(L/D)_{max}$	$(L/D)_{max}$
Prop	$(L/D)_{max}$	$0.866\,(L/D)_{max}$

- For initial sizing, these percentages should be used to estimate L/D from $(L/D)\text{max}$

Cruise

$$\frac{W_i}{W_{i-1}} = e^{\frac{-RC}{V(L/D)}}$$

Loiter

$$\frac{W_i}{W_{i-1}} = e^{\frac{-EC}{L/D}}$$

$$\frac{W_{final}}{W_{initial}} = \overbrace{\frac{W_1}{W_0}}^{\text{Takeoff}}\overbrace{\frac{W_2}{W_1}}^{\text{Climb}}\overbrace{\frac{W_3}{W_2}}^{\text{Cruise}}\overbrace{\frac{W_4}{W_3}}^{\text{Descend}}\overbrace{\frac{W_5}{W_4}}^{\text{Loiter}}\overbrace{\frac{W_6}{W_5}}^{\text{Descend}}\overbrace{\frac{W_7}{W_6}}^{\text{Landing}}$$

$$W_{initial} - W_{final} = W_{fuel}$$

$$1 - \frac{W_{final}}{W_{initial}} = \frac{W_{fuel}}{W_{initial}}$$

Fuel-fraction Estimation

$$W_{\text{initial}} - W_{final} = W_{fuel}$$

Mission fuel fraction:

$$1 - \frac{W_{\text{final}}}{W_{initial}} = 1 - \frac{W_x}{W_0}$$

- In general: **6% reserve and trapped fuel allowance:**

$$\frac{W_f}{W_0} = 1.06\left(1 - \frac{W_x}{W_0}\right)$$